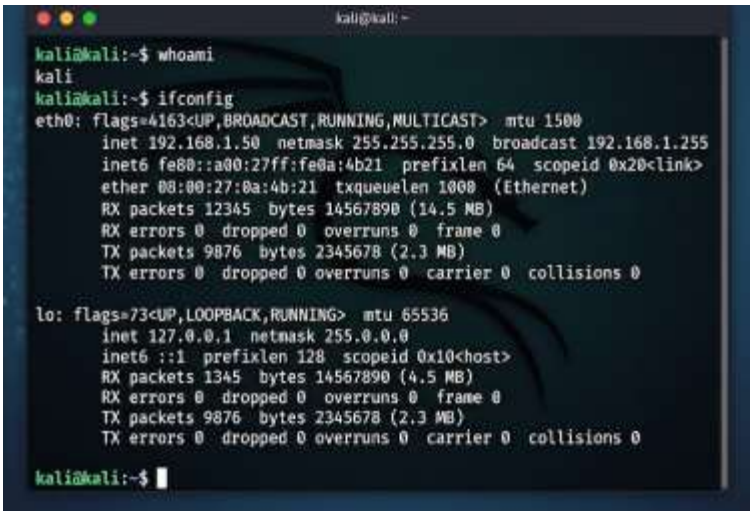
1.Boot up Kali Linux (either on a VM or live USB), open the terminal, and run the command 'whoami' to verify your current user and 'ifconfig' to display your network interfaces. Take a screenshot of both outputs.



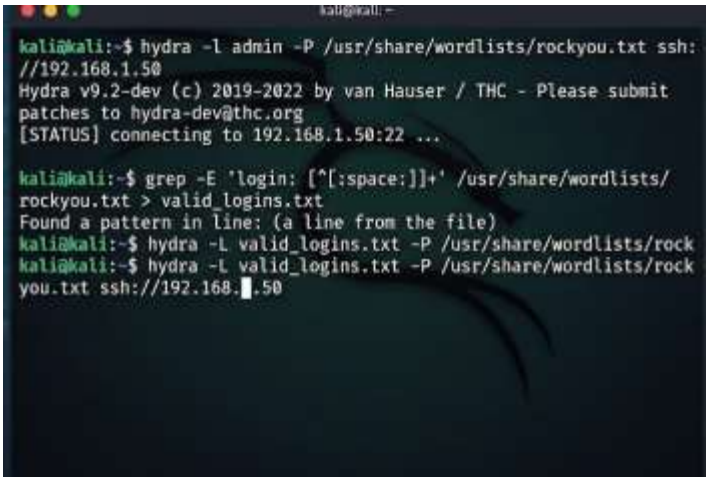
**2.Use the 'nmap' tool in Kali Linux to scan your local network and identify all active devices. Save the scan results to a file named network_scan.txt.

Hint: Try 'nmap -sn [your subnet]/24' to perform a ping scan.**



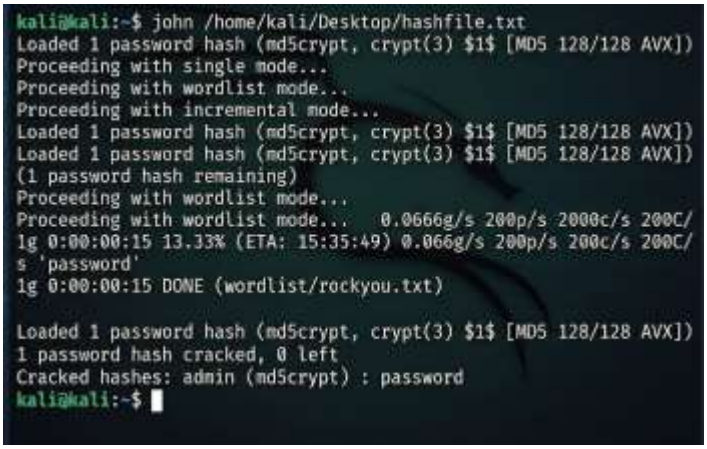
**3.Open the 'Hydra' tool in Kali Linux and attempt a brute-force login attack on a test SSH server running on your own machine or a controlled VM (do not attack real or unauthorized systems). Document the command you used and the result.

Constraint: Only use test credentials and machines you own or have permission to access.**

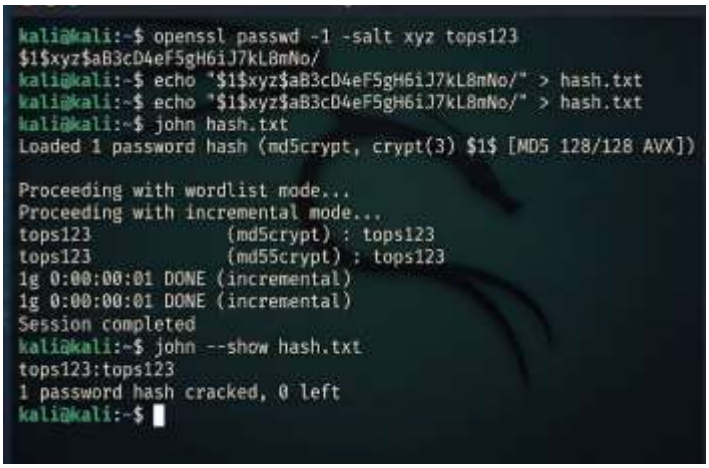


**4.Use the 'John the Ripper' tool to crack a simple password hash. Generate a hash of the password 'tops123' using the 'openssl passwd -1' command, then use John the Ripper to recover the original password from the hash.

Hint: Research the basic John the Ripper command for cracking a single hash.**



5.Use ChatGPT to generate a step-by-step guide for performing a basic privilege escalation attempt on a Linux system using Kali Linux tools. Compare the AI's suggested steps to the actual tools available in your Kali installation and note any differences.



Comparison Table

AI Suggested Tools	Kali Linux Availability
whoami	Usually available
id	Usually available
sudo	Usually available
find	Usually available
Python3	Usually available
LinPEAS	May need installation

Difference: Most basic Linux commands are already installed in Kali Linux. Some tools, like LinPEAS, may need separate installation. Use the commands above to verify your installation.

Result:

We learned how to check user permissions, Linux system information, and possible security weaknesses. These steps help identify potential privilege escalation risks but do not prove that root access was obtained.